



SENIOR THESIS IN MATHEMATICS

Exploration of the Gaussian Mixture Model and Its Application to Baseball Analytics

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Abstract

Within this paper, we answer the question: What is the Gaussian mixture model and how can it be used to analyze MLB bat tracking and pitching data? Additionally, we will be diving deep into the covariance structures that develop the numerous clustering models. In order for us to understand the accuracy of different models, we will simulate data sets built off of specific covariance structures and then run differing clustering models to see how the model performs. Lastly, we will apply our understanding of the Gaussian mixture model to MLB bat tracking and pitching data to draw analysis on various facets of the game.

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Chapter 1

Introduction

From the turn of the century, professional baseball has spearheaded the drive towards revolutionizing sports analytics. In this field, the revolution was first exposed with the release of *Moneyball* (Lewis, 2003), a book that explored winning in the MLB as a low-budget franchise using data-driven insights. In particular, the book put an emphasis on the importance of utilizing undervalued metrics to inform decisions that will benefit the team. The release of *Moneyball* sparked a sport-wide push into the data analytics market showing rapid growth of analytics departments within many franchises. Baseball analysis was dominated by qualitative observations from scouts and coaches for decades. Now, with a data-driven perspective on the sport, qualitative observations can be paired with the quantitative to enhance the understanding of a player's value to a team.

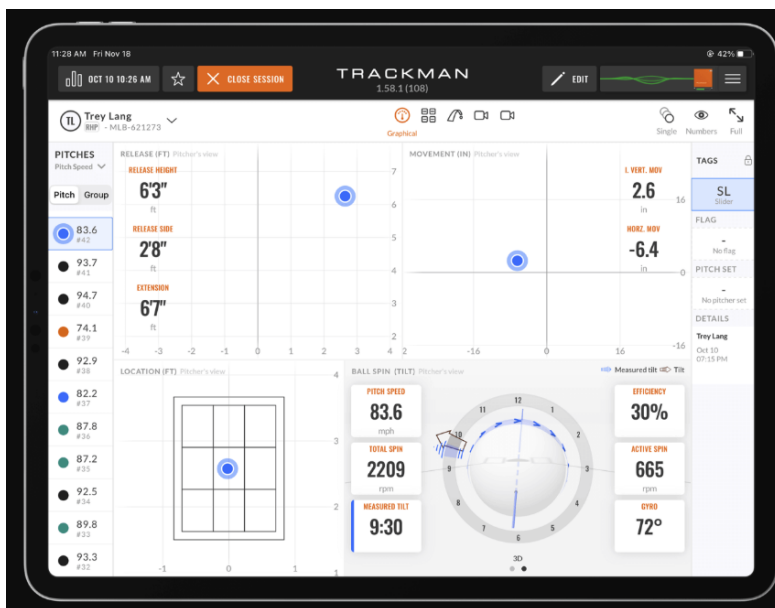


Figure 1.1: Sample image of a single pitch Trackman output (Trackman, 2023)

After decades of advancement within the field, baseball analytics has stifled its growth, embracing the metrics that have influenced the game for decades. With this stifling, professional baseball has shifted their focus towards advancing the game technologically. One piece of tech that has primarily benefited pitchers being Trackman, a radar technology that tracks characteristics of baseball in-motion and has helped player development exponentially for over a decade (see Figure 1.1). On the batter side, we have seen growth in the development of bat tracking technology. Bat tracking technology gives detailed information on the characteristics of a batter's stance and swing, allowing them to derive and analyze their strengths and weaknesses through biometrics. Although this technology can be applied to batter specific instances, baseball analytics databases like BaseballSavant (2024) has used the Trackman technology to develop new bat tracking metrics.

Midway through the 2023 MLB season, BaseballSavant (2024) introduced the world of baseball analytics to five new bat tracking metrics: attack angle, attack direction, swing path tilt, swing length, and bat speed. Figures 1.2, 1.3, and 1.4 represent the metrics with a biometric skeleton followed by the definition.

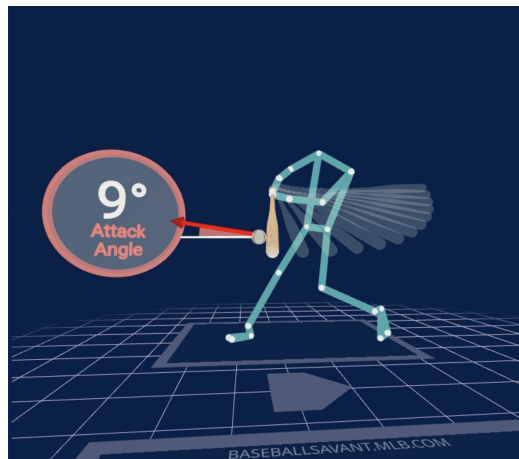


Figure 1.2: **Attack angle** is defined as - The vertical angle at which the sweet spot of the bat (an approximate area on a bat that results in optimized ball contact) is traveling at the point of impact with the ball. (BaseballSavant, 2024)

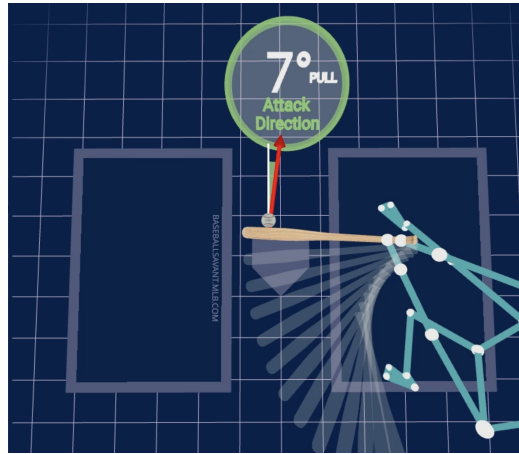


Figure 1.3: **Attack direction** is defined as - The horizontal angle at which the sweet spot of the bat is traveling at the point of impact with the ball, expressed as an angle to the “PULL” or “OPPO” direction. (BaseballSavant, 2024)

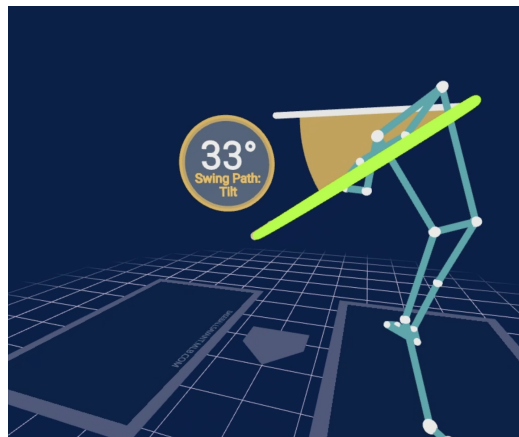


Figure 1.4: **Swing path tilt** is defined as - The vertical angle of the arc traced by the swing path over the 40 ms prior to contact. A higher tilt indicates a “steeper” swing, while a lower tilt indicates a “flatter” swing. (BaseballSavant, 2024)

The fourth metric, **bat speed**, is defined as - the speed of the bat measured at the sweet spot of the bat. One MLB (2025) defines sweet spot as: a classification [that] quantifies a batted-ball event with a launch angle ranging from 8 to 32. The final metric, **swing length**, is defined as - the total (sum) distance in feet traveled of the head of the bat in X/Y/Z space, from start of tracking data until impact point. (BaseballSavant, 2024)

Before the release of the Trackman metrics, a batter’s swing was limited to qualitative descriptions. With the release of these five new metrics, the baseball world now has the capability to mathematically describe a batter’s swing. Rather than just stating that a batter’s swing is flat, an analyst can now refer to the exact angle in which said batter’s “flatness” resides.

A question that has puzzled the baseball community is whether there exists multiple types of swings within a batter’s arsenal. Players having multiple swing types within their arsenal have always existed through concepts like a “two-strike approach” (which refers to a batter manipulating their swing to prioritize making contact and consequently reducing the probability of striking out). When two-time MVP, eight-time all star, and four-time silver slugger Bryce Harper was asked by DrivelineBaseball (2025) how his swing has changed over the years, he responded by explaining that he now possesses multiple swings with quote:

“I think for me it’s [that] I have more clubs in the bag... there are times when I do a toe tap, times where I do leg kick, and times where I do a two-strike approach... I am one of the ones in the game who changes their swing a lot from one day to another.”

Reading a quote from such a decorated player within the game’s history raises the question: Do different swings exist within the profile of multiple players? By combining the swing-by-swing data entries of the Baseball Savant data with Gaussian mixture models gives us the opportunity to explore batting profiles. My research is to explore whether there is statistical evidence for batters having distinct swings by using Gaussian mixture techniques.

Chapter 2

Mixture Distributions

Within this chapter, we will be investigating mixture distributions, and the individual elements that comprise them. The information in this chapter was gathered from (Scrucca et al., 2023) which discusses a systematic statistical approach to clustering data.

Definition 2.1. A mixture distribution is a probability distribution obtained as a convex linear combination of probability density functions.(Scrucca et al., 2023)

The general form of the density of a finite mixture distribution:

$$f(x; \theta) = \sum_{k=1}^G \pi_k f_k(x; \theta_k) \quad (2.1)$$

where:

- G represents the number of mixture components.
- π_k is the parameter that describes the total proportion of the distribution being attributed to the k^{th} component ($\pi_k > 0$ and $\sum_{k=1}^G \pi_k = 1$).
- θ_k represents the set of parameters of the k^{th} component's probability distribution.
- f_k is the k_{th} probability function that maps the overall density of the observations.

Mixture distributions can be used to represent an array of phenomena within statistics. One prominent phenomena being the existence of unobserved heterogeneity.

Definition 2.2. Closer (2025) defines **unobserved heterogeneity** as the existence of unmeasured (unobserved) differences between study participants or samples that are associated with the (observed) variables of interest.

In other words, unobserved heterogeneity occurs when a sample is taken from a population, without information about the sub-populations that exist within the extracted sample. To illustrate this concept of unobserved phenomena, we will construct probability distributions for an arbitrary data set that partitions the data into $G = 1, 2, 3$ component(s).

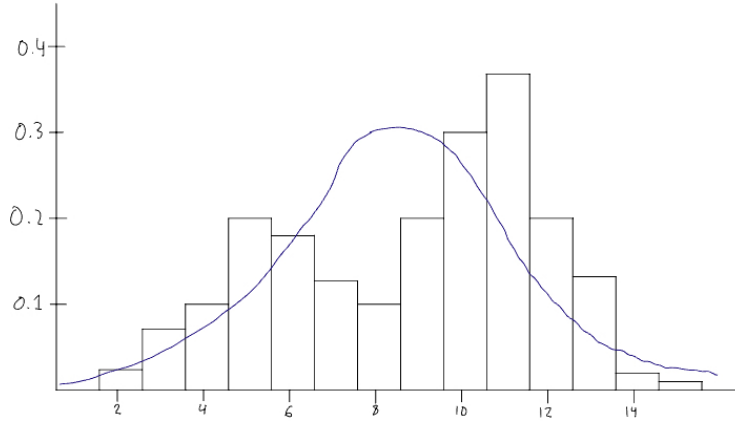


Figure 2.1: Probability distribution curve when fit with $G = 1$ component.

As seen in Figure 2.1, we let $G = 1$ and construct a unimodal Gaussian curve to indicate the distribution of the data, assuming this curve to be normally distributed. Now what happens when $G = 2$ components?

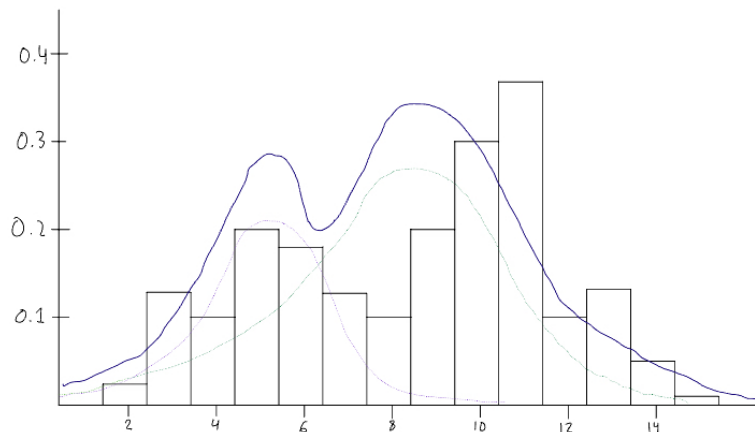


Figure 2.2: Probability distribution curve when fit with $G = 2$ components.

In Figure 2.2, we now estimate a bimodal curve (solid curve) and two Gaussian curves (dashed curves) from the data set to indicate a partitioning of the data into two distinct subgroups. The dashed curves represent the probability distribution of the partitioned data. It is also very important to note that with these new probability functions, there exists distinct μ_1, μ_2 and σ_1, σ_2 for the two probability density curves. Now, what happens when $G = 3$ components?

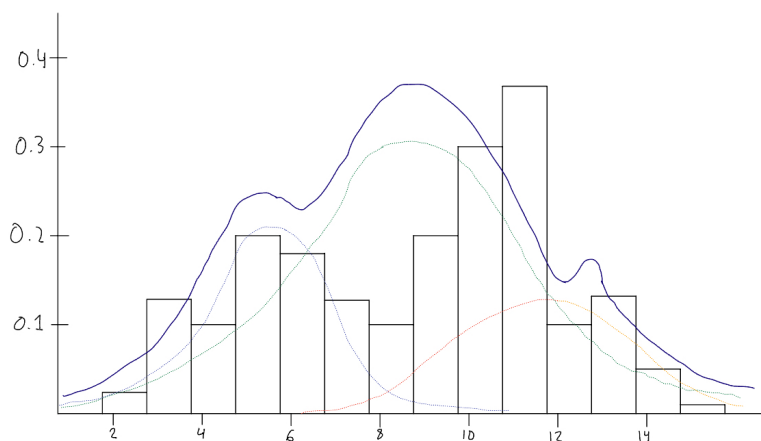


Figure 2.3: Probability distribution when fit with $G = 3$ components.

In Figure 2.3, now that $G = 3$ we extend the logic from Figure 2.2 to

construct a trimodal probability distribution curve (solid curve) and three normal probability distribution curves (dashed curves) to represent the partitioning of data into $G = 3$ components.

Note: Figures 2.1, 2.2, and 2.3 seek to describe a distribution with an unknown number of components. The figures are simply used as a representation of how increasing/decreasing the number of components G influences estimation of the mixture distribution. Additionally, these figures do not represent the fact that $\sum_{k=1}^G \pi_k f_k(x; \theta_k) = 1$.

After extending our partition into $G = 3$ components, we are estimating subgroups that could exist within the data set and hold observational relevance thus resulting in $G = 3$ clusters. But rather than just constructing arbitrary curves within the data, how do we extract relevant parameters that describe properties of the curves? That is, how do we actually find the appropriate parameters associated with each cluster? Additionally, how do we know which data points belong to which “correct” cluster? Essentially, when we are clustering a data set, we need to allocate each point to their “correct” cluster, and then calculate the parameters for each cluster. Thankfully, we can turn to the Expectation-Maximization (EM) algorithm to answer these questions.

Chapter 3

The Expectation - Maximization (EM) Algorithm

If we were given a data set and tasked with clustering a data set into three groups, as accurately as possible, how would we do it? Let's visualize this scenario:

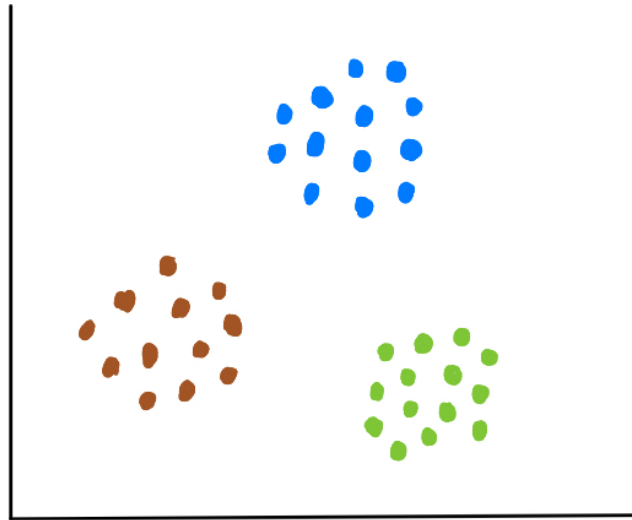


Figure 3.1: Scenario 1

In Scenario 1, it's an easy task, we can cluster the data by eye. Addition-

ally, we can easily derive θ_k parameters as well by calculations (assuming we have the values associated with each data point).

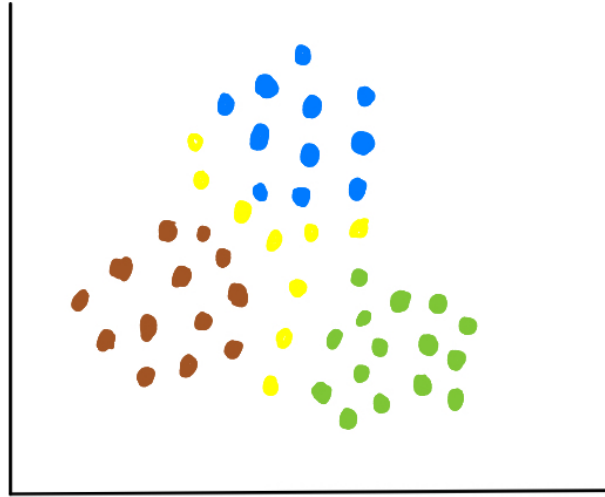


Figure 3.2: Scenario 2

Figure 3.2 shows four different populations, however, the yellow observations are not as distinguishable as the previous three clusters. Now, the cluster groupings are not as easy to distinguish by eye. But, we can still do our best to cluster the data and then calculate θ set of parameters. **Note:** when analyzing Scenario 2, how should we classify the unclustered (yellow) points? Should they be labeled as blue points or should it be labeled as green points?

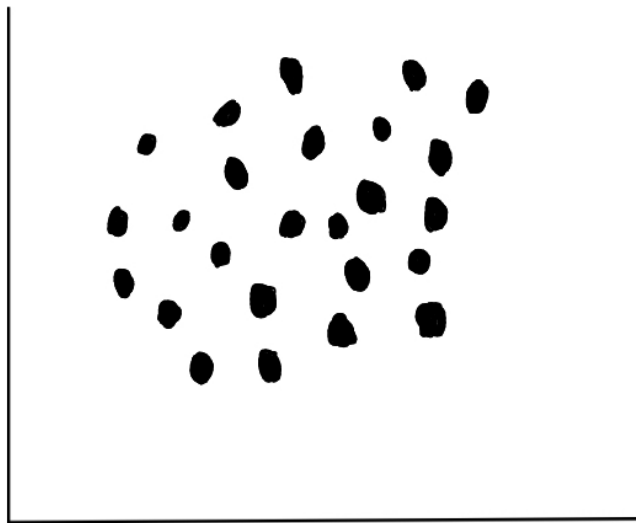


Figure 3.3: Scenario 3

Scenario 3 causes us some trouble because we don't have any idea about how to partition the observations. Without any mathematical method, clustering real world data is essentially an impossible task. Additionally, for Scenarios 2 and 3, we would be calculating θ_k incorrectly because we allocated the data points using an arbitrary method, leaving substantial room for error. Luckily, there exists an algorithm to save the day and solve this problem: The Expectation-Maximization (EM) algorithm.

The EM algorithm uses maximum likelihood estimation (MLE) to assess how accurate the cluster estimates are. Additionally, since the data points' cluster allocation is variable, this process must be done several times until we are left with parameters that result in the largest log-likelihood. **Note:** within the context of this paper, the “missing-data” that is mentioned refers to the assignment of an observation x_i to the correct component k . A general way of understanding the EM algorithm's iterative process is as follows:

1. Obtain initial estimates of the missing data.
2. Calculate the maximum likelihood estimate (MLE) for each component

of interest as if the missing data had been found.

3. Calculate new estimates of the missing data.
4. Repeat steps 2 and 3 until both the MLE and the missing data estimates converge.

Definition 3.1. The Expectation-Maximization (EM) algorithm is an iterative procedure whose objective function at each iteration is the conditional expectation of the complete-data log-likelihood, the Q-function, which for finite mixtures takes the form:

$$Q(\Psi; \Psi^{(t)}) = \sum_{i=1}^n \sum_{k=1}^G \hat{z}_{ik}^{(t)} \{ \log \pi_k + \log f_k(x_i; \theta_k) \} \quad (3.1)$$

Where the log-likelihood (using Gaussian parameters) is defined as:

$$\ell(\Psi) = \sum_{i=1}^n \sum_{k=1}^G \hat{z}_{ik} \{ \log \pi_k + \log \left[\frac{1}{\sqrt{2\pi\sigma_k^2}} e^{-\frac{(x_i - \mu_k)^2}{2\sigma_k^2}} \right] \} \quad (3.2)$$

Note: $\Psi = (\mu_1, \pi_1, \sigma_1, \dots, \mu_G, \pi_G, \sigma_G)$ set of parameters and θ_k is the subset of parameters associated with k cluster ($\theta_1 = \mu_1, \pi_1, \sigma_1$). Moreover, $\hat{z}_{ik}^{(t)} = E(z_{ik} = 1 | x_i, \Psi^{(t)})$, the estimated conditional probability that x_i belongs to the k th component at iteration t of the algorithm is defined as:

$$\hat{z}_{ik}^{(t)} = \mathbb{P}(z_{ik} = 1 | x_i, \hat{\Psi}^{(t)}) = \frac{\pi_k^{(t)} f_k(x_i; \theta_k^{(t)})}{\sum_{j=1}^G \pi_j^{(t)} f_j(x_i; \theta_j^{(t)})} \quad (3.3)$$

The Q-function takes a similar form as the likelihood equation 2.1, except now we take the $\log$, and also multiply by $\hat{z}_{ik}^{(t)}$ which is defined above. Then, by estimating the likelihood that observation x_i belongs to a given component k . Once the EM algorithm has been run iteratively t times, we then get the optimized likelihood parameters θ_k that now makes-up the “complete-data”.

3.0.1 Likelihood Maximization

For this iterative process, we need to maximize the likelihood of the data appropriately. Subsequently, the most appropriate course to take with maximizing the likelihood would be finding $\mu_{k'}$ when: $\frac{\partial \ell(\Psi)}{\partial \mu_{k'}} = 0$, $\pi_{k'}$ when: $\frac{\partial \ell(\Psi)}{\partial \pi_{k'}} = 0$, and σ_k when: $\frac{\partial \ell(\Psi)}{\partial \sigma_{k'}} = 0$.

Note: for this procedure, since we are finding the sets of θ_k parameters that maximize the likelihood we will be using “ k' ” to indicate the procedure being done for one arbitrary set of $\theta_{k'}$ parameters.

For $\mu_{k'}$ when $\frac{\partial \ell(\Psi)}{\partial \mu_{k'}} = 0$ we will show that the parameter $\frac{\sum_{i=1}^n z_{ik'} \cdot x_i}{\sum_{i=1}^n z_{ik'}}$ maximizes the likelihood:

$$\frac{\partial \ell(\Psi)}{\partial \mu_k} = \frac{\partial}{\partial \mu_k} \left(\sum_{i=1}^n \sum_{k=1}^G z_{ik} \{ \ln(\pi_k) + \ln \left[\frac{1}{\sqrt{2\pi\sigma_k^2}} \cdot e^{-\frac{(x_i - \mu_k)^2}{2\sigma_k^2}} \right] \} \right)$$

To begin the derivation, by linearity of differentiation, we can rewrite $\frac{\partial \ell(\Psi)}{\partial \mu_k}$ as:

$$0 = \sum_{i=1}^n \frac{\partial}{\partial \mu_{k'}} (z_{ik'} \cdot \ln(\pi_{k'})) + \frac{\partial}{\partial \mu_{k'}} (z_{ik'} \cdot \ln \left[\frac{1}{\sqrt{2\pi\sigma_{k'}^2}} \cdot e^{-\frac{(x_i - \mu_{k'})^2}{2\sigma_{k'}^2}} \right])$$

Since $z_{ik'}$ is a constant, with respect to $\mu_{k'}$, we can distribute it out of the derivative. Moreover, since we are deriving with respect to $\mu_{k'}$, $\ln(\pi_{k'})$ and $\frac{1}{\sigma_{k'}^2}$ are treated as constants canceling out the values. Additionally, by distributing $\ln$ to the quantities, e cancels out to get:

$$0 = \sum_{i=1}^n z_{ik'} \cdot \frac{\partial}{\partial \mu_{k'}} \left(\ln \left[\frac{1}{\sqrt{2\pi\sigma_{k'}^2}} \right] - \frac{(x_i - \mu_{k'})^2}{2\sigma_{k'}^2} \right)$$

Now, since we are deriving with respect to $\mu_{k'}$, $\ln \left[\frac{1}{\sqrt{2\pi\sigma_{k'}^2}} \right]$ is treated as a constant giving us:

$$0 = \sum_{i=1}^n z_{ik'} \cdot \frac{\partial}{\partial \mu_{k'}} \left(-\frac{(x_i - \mu_{k'})^2}{2\sigma_{k'}^2} \right)$$

Distribute constant $\frac{1}{2}$ out of the partial:

$$0 = \sum_{i=1}^n z_{ik'} \cdot -\frac{1}{2} \cdot \frac{\partial}{\partial \mu_{k'}} (x_i - \mu_{k'})^2$$

By deriving $(x_i - \mu_{k'})^2$ using the chain rule, we now have:

$$\begin{aligned} 0 &= \sum_{i=1}^n z_{ik'} \cdot (x_i - \mu_{k'}) \\ 0 &= \sum_{i=1}^n z_{ik'} \cdot x_i - \sum_{i=1}^n z_{ik'} \cdot \mu_{k'} \end{aligned}$$

Next, by subtracting $\sum_{i=1}^n z_{ik'} \cdot \mu_{k'}$ we get:

$$\sum_{i=1}^n z_{ik'} \cdot \mu_{k'} = \sum_{i=1}^n z_{ik'} \cdot x_i$$

Finally, divide each side by $\sum_{i=1}^n z_{ik'}$ to get:

$$\hat{\mu}_{k'}^{(t)} = \frac{\sum_{i=1}^n z_{ik'} \cdot x_i}{\sum_{i=1}^n z_{ik'}} \quad (3.4)$$

For $\pi_{k'}$ when $\frac{\partial \ell(\Psi)}{\partial \pi_{k'}} = 0$, we will show that the parameter $\pi_{k'} = \frac{n_{k'}}{\pi_{k'}}$ maximizes the likelihood:

$$\frac{\partial \ell(\Psi)}{\partial \pi_k} = \frac{\partial}{\partial \pi_{k'}} \left(\sum_{i=1}^n \sum_{k=1}^G z_{ik} \{ \ln(\pi_k) + \ln \left[\frac{1}{\sqrt{2\pi\sigma_k^2}} e^{-\frac{(x_i - \mu_k)^2}{2\sigma_k^2}} \right] \} \right)$$

Note: For $\pi_{k'}$, since there is a constraint of $\sum_k \pi_k = 1$, it is necessary for us to apply λ as the lagrange multiplier and then solve. Moreover, by linearity of differentiation, we can rewrite the equation as:

$$0 = \sum_{i=1}^n z_{ik'} \frac{\partial}{\partial \pi_{k'}} (\ln(\pi_{k'})) + \frac{\partial}{\partial \pi_{k'}} \left(\ln \left[\frac{1}{\sqrt{2\pi\sigma_{k'}^2}} e^{-\frac{(x_i - \mu_{k'})^2}{2\sigma_{k'}^2}} \right] \right) + \frac{\partial}{\partial \pi_{k'}} \left(\lambda \left(\sum_{k=1}^G \pi_k - 1 \right) \right)$$

Since we are computing the derivative with respect to $\pi_{k'}$, $\ln[\frac{1}{\sqrt{2\pi\sigma_{k'}^2}}e^{-\frac{(x_i-\mu_{k'})^2}{2\sigma_{k'}^2}}]$ is treated as a constant, thus making $\frac{\partial}{\partial\pi_{k'}}(\ln[\frac{1}{\sqrt{2\pi\sigma_{k'}^2}}e^{-\frac{(x_i-\mu_{k'})^2}{2\sigma_{k'}^2}}]) = 0$ resulting in:

$$0 = \sum_{i=1}^n z_{ik'} \frac{\partial}{\partial\pi_{k'}} (\ln(\pi_{k'})) - \lambda$$

$$0 = \sum_{i=1}^n z_{ik'} \frac{1}{\pi_{k'}} - \lambda$$

Add λ to both sides:

$$\lambda = \frac{\sum_{i=1}^n z_{ik'}}{\pi_{k'}}$$

Finally, let $n_{k'} = \sum_{i=1}^n z_{ik'}$ and multiply both sides by $\frac{\pi_{k'}}{\lambda}$:

$$\hat{\pi}_{k'}^{(t)} = \frac{n_{k'}}{\lambda} \quad (3.5)$$

For $\sigma_{k'}$ when $\frac{\partial\ell(\psi)}{\partial\sigma_k} = 0$, we will show that the parameter $\sigma_{k'} = \sqrt{\frac{\sum_{i=1}^n z_{ik'}(x_i-\mu_{k'})^2}{\sum_{i=1}^n z_{ik'}}}$, maximizes the likelihood:

$$\frac{\partial\ell(\Psi)}{\partial\sigma_k} = \frac{\partial}{\partial\sigma_k} (\sum_{i=1}^n \sum_{k=1}^G z_{ik} \{ \ln(\pi_k) + \ln[\frac{1}{\sqrt{2\pi\sigma_k^2}} e^{-\frac{(x_i-\mu_k)^2}{2\sigma_k^2}}] \})$$

To begin the derivation, by linearity of differentiation, we can rewrite the equation as:

$$0 = \sum_{i=1}^n z_{ik'} \frac{\partial}{\partial\sigma_{k'}} (\ln(\pi_{k'})) + \frac{\partial}{\partial\sigma_{k'}} (\ln[\frac{1}{\sqrt{2\pi\sigma_{k'}^2}}]) - \frac{\partial}{\partial\sigma_{k'}} (\frac{(x_i - \mu_{k'})^2}{2\sigma_{k'}^2})$$

Now, since the derivative is with respect to $\sigma_{k'}$, $\frac{\partial}{\partial \sigma_{k'}}(\ln(\pi_{k'})) = 0$ and then compute the partial derivatives as follows:

$$0 = \sum_{i=1}^n z_{ik'} \left(\frac{\partial}{\partial \sigma_{k'}} (-\ln[\sqrt{2\pi\sigma_{k'}^2}]) - \frac{\partial}{\partial \sigma_{k'}} \left(\frac{(x_i - \mu_{k'})^2}{2\sigma_{k'}^2} \right) \right)$$

After applying the quotient rule to $\frac{(x_i - \mu_{k'})^2}{2\sigma_{k'}^2}$ and computing $\frac{\partial}{\partial \sigma_{k'}}[\ln(\sqrt{2\pi\sigma_{k'}^2})]$, $\frac{\partial \ell(\psi)}{\partial \sigma_{k'}}$ can be evaluated as follows:

$$\begin{aligned} 0 &= \sum_{i=1}^n z_{ik'} \left[-\frac{1}{2} \left(\frac{4\pi\sigma_{k'}}{2\pi\sigma_{k'}^2} \right) + \frac{(x_i - \mu_{k'})^2}{\sigma_{k'}^3} \right] \\ 0 &= \sum_{i=1}^n z_{ik'} \left[\frac{-1}{\sigma_{k'}} + \frac{(x_i - \mu_{k'})^2}{\sigma_{k'}^3} \right] \end{aligned}$$

By the distributive property, $\sum_{i=1}^n z_{ik'}$ can be distributed to both terms to result in:

$$0 = -\sum_{i=1}^n \frac{z_{ik'}}{\sigma_{k'}} + \sum_{i=1}^n \frac{z_{ik'}(x_i - \mu_{k'})^2}{\sigma_{k'}^3}$$

Add $\sum_{i=1}^n \frac{z_{ik'}}{\sigma_{k'}}$ to both sides:

$$\sum_{i=1}^n \frac{z_{ik'}}{\sigma_{k'}} = \sum_{i=1}^n \frac{z_{ik'}(x_i - \mu_{k'})^2}{\sigma_{k'}^3}$$

Now, we can multiply $\frac{\sigma_{k'}^3}{\sum_{i=1}^n z_{ik'}}$ to each side allowing terms to cancel out resulting in the equation:

$$\begin{aligned} \frac{\sigma_{k'}^3}{\sum_{i=1}^n z_{ik'}} \cdot \frac{\sum_{i=1}^n z_{ik'}}{\sigma_{k'}} &= \frac{\sum_{i=1}^n z_{ik'}(x_i - \mu_{k'})^2}{\sigma_{k'}^3} \cdot \frac{\sigma_{k'}^3}{\sum_{i=1}^n z_{ik'}} \\ \sigma_{k'}^2 &= \frac{\sum_{i=1}^n z_{ik'}(x_i - \mu_{k'})^2}{\sum_{i=1}^n z_{ik'}} \end{aligned}$$

Finally, after taking the square root of each side we are left with:

$$\sigma_{k'}^{(t)} = \sqrt{\frac{\sum_{i=1}^n z_{ik'} (x_i - \mu_{k'})^2}{\sum_{i=1}^n z_{ik'}}} \quad (3.6)$$

Now, that we have shown that $\hat{\mu}_{k'}^{(t)} = \frac{\sum_{i=1}^n z_{ik'} x_i}{\sum_{i=1}^n z_{ik'}}$, $\hat{\pi}_{k'} = \frac{n_{k'}}{\lambda}$, and $\hat{\sigma}_{k'} = \sqrt{\frac{\sum_{i=1}^n z_{ik'} (x_i - \mu_{k'})^2}{\sum_{i=1}^n z_{ik'}}}$, what does this mean? Going back to Figures 3.1, 3.2, and 3.3, we can now find the θ_k parameters that maximize the likelihood! Now that we understand how the EM algorithm calculates the “missing” parameters we can understand how the EM algorithm works for a k component:

1. Define the initial estimate of π_k and θ_k as $\pi_k^{(0)}$ and $\theta_k^{(0)}$.
2. E-Step: Estimate the latent component memberships using Equation 3.3
3. M-Step: Plug Equation 3.3 into Equations 3.4, 3.5, and 3.6 to obtain updated parameter estimates.
4. If “convergence” is not met, set $t = t + 1$ and perform steps 2 and 3 again

Note: “convergence” is difficult to assess and is essentially subjective. “Convergence” can be interpreted as the parameters only changing a little after each $t + 1$ iteration.

Chapter 4

Multivariate Clustering

Within this chapter (and subsequent chapters) we will be analyzing mixture distributions through a multivariate lens.

Covariance matrices are an essential element of the data analysis process. For each covariance structure, the geometric structure is solely influenced by covariance matrices through eigen-decomposition:

$$\sum_k = \lambda_k U_k \Delta_k U_k^\top, \quad (4.1)$$

λ_k is a scalar that controls the volume of the cluster. For clustering models, VII, VEI, VVI, VEE, VVE, VEV, and VVV λ_k is a scalar that is unique to each k cluster. On the contrary, in clustering models EII, EEI, EVI, EEE, EVE, EEV, and EVV λ_k is instead denoted as λ to define λ as a single scalar multiplied over all k number of clusters. Δ_k is a diagonal matrix controlling the shape of the cluster. Each value in the diagonal matrix corresponds to a specific dimension being altered. U_k (and subsequently $U_k^\top$) are orthogonal matrices of eigenvectors that control the orientation of the covariance structure. Similarly to both λ_k and U_k can either be singular matrices that can be used for each k clusters or be k unique matrices that are applied to each cluster. (Scrucca et al., 2023)

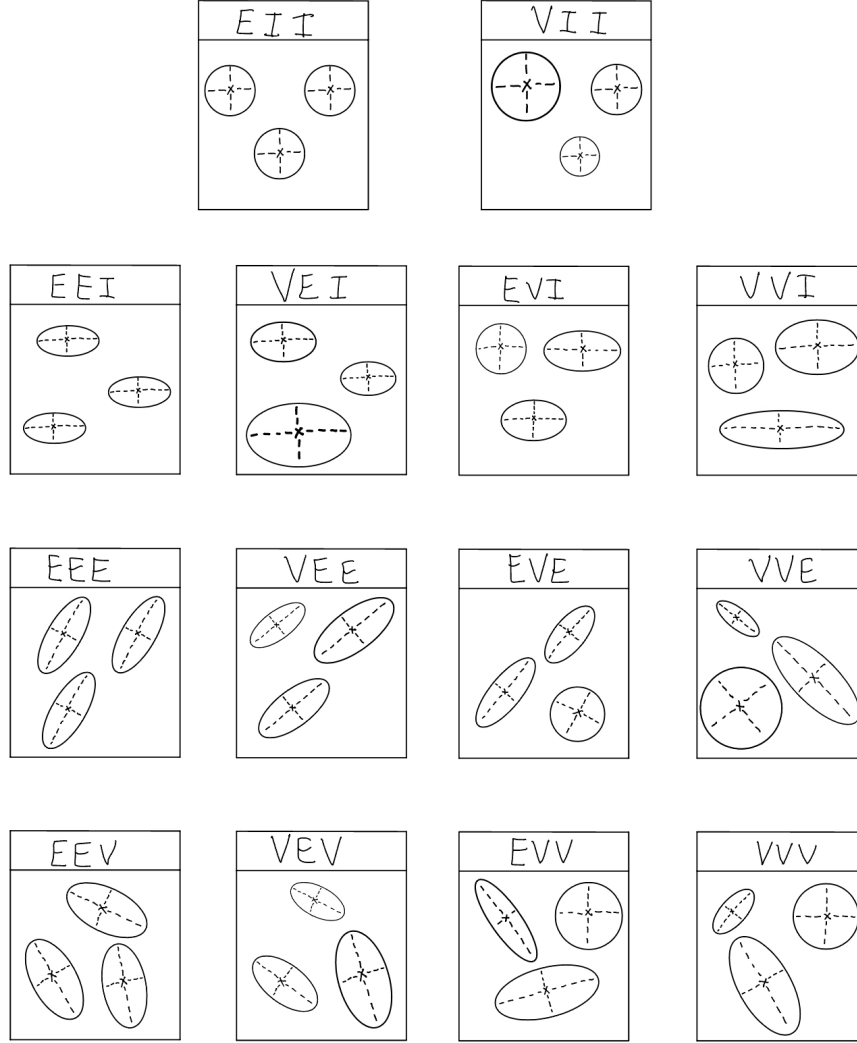


Figure 4.1: These figures represent the 14 possible Gaussian models with $G = 3$ components.

In Figure 4.1 the first row of models (characterized by their “-II” labeling indicating U and Δ being equal to the identity matrix) represent spherical covariance structures parameterized by λ/λ_k . In these two models, we see how λ is the parameter that controls the size of the cluster. The second row of figures (characterized by their “-I” labeling indicating U being equal to

the identity matrix) represent diagonal covariance structures parameterized by λ/λ_k and Δ/Δ_k . Clusters under this categorization can simultaneously have their volumetric size and shape controlled by these parameters. The third row of figures (characterized by their “-E” labeling) represent models of equal orientation parameterized by λ/λ_k , Δ/Δ_k , and $UU^\top$. Meaning, these models can form clusters of similar identity to row two, however now we can set a distinct orientation for the clusters. Lastly, the fourth row of models (characterized by their -V labeling) represent models of varying-orientation parameterized by λ/λ_k , Δ/Δ_k , and $UU^\top / U_k U_k^\top$. Scrucca et al. (2023)

Label	Model	Number of parameters
EII	$\lambda \mathbf{I}$	$(G - 1) + Gd + 1$
VII	$\lambda_k \mathbf{I}$	$(G - 1) + Gd + G$
EEI	$\lambda \Delta$	$(G - 1) + Gd + d$
VEI	$\lambda_k \Delta$	$(G - 1) + Gd + G + (d - 1)$
EVI	$\lambda \Delta_k$	$(G - 1) + Gd + 1 + G(d - 1)$
VVI	$\lambda_k \Delta_k$	$(G - 1) + Gd + G + G(d - 1)$
EEE	$\lambda U \Delta U^\top$	$(G - 1) + Gd + 1 + (d - 1) + d(d - 1)/2$
VEE	$\lambda_k U \Delta U^\top$	$(G - 1) + Gd + G + (d - 1) + d(d - 1)/2$
EVE	$\lambda U \Delta_k U^\top$	$(G - 1) + Gd + 1 + G(d - 1) + d(d - 1)/2$
VVE	$\lambda_k U \Delta_k U^\top$	$(G - 1) + Gd + G + G(d - 1) + d(d - 1)/2$
EEV	$\lambda U_k \Delta U_k^\top$	$(G - 1) + Gd + 1 + (d - 1) + Gd(d - 1)/2$
VEV	$\lambda_k U_k \Delta U_k^\top$	$(G - 1) + Gd + G + (d - 1) + Gd(d - 1)/2$
EVV	$\lambda U_k \Delta_k U_k^\top$	$(G - 1) + Gd + 1 + G(d - 1) + Gd(d - 1)/2$
VVV	$\lambda_k U_k \Delta_k U_k^\top$	$(G - 1) + Gd + G + G(d - 1) + Gd(d - 1)/2$

Figure 4.2: Number of parameters being estimated for each model.(Scrucca et al., 2023)

In Figure 4.2, the *Label* column defines how each clustering model is labeled. The *Model* column classifies which parameters are included in each clustering model. Finally, the *Number of parameters* column shows the equation for the number of parameters in the model (given G = the number of components and d = the dimension of the dataset).

4.1 Simulation Analysis

For our simulation analysis, we will investigate whether there are certain models that recover information more accurately. What does this mean? Well, imagine a data set that has unobserved homogeneity. If we run a clustering model under the “VVV” covariance model, then the resulting classification output will overfit the data because of the large number of parameters being optimized. This idea is seen in Figure 4.2 where the number of parameters increases drastically as we go down the list of models. What if we run the data under the “EEE” covariance model? Then, the resulting classification output will result in an extremely under fit clustering, due to an insufficient amount of parameters to accurately fit the data. So the question is, how do we find a middle ground?

below are the following steps for our simulation analysis:

1. Choose a covariance model to build the structure our data .
2. Choose the λ/λ_k , Δ/Δ_k , and/or $UU^\top / U_kU_k^\top$ parameters.
3. Create a Gaussian distributed 900 entry three-dimensional data set following the chosen parameters.
4. Cluster the data under the “wrong” model and also under the “correct” (one clustering test under the EEE model and then another test under the chosen model).
5. Compute the adjusted rand index (ARI) for the simulated data.
6. Repeat 500 times.

Covariance Matrix: For this experiment we needed to create a function that can form the correct covariance matrix when a specific model is the input.

Parameters:

- λ/λ_k is defined as a length one or three list of scalars.
- Δ/Δ_k is defined as a length one (or three) list of length three diagonal matrices.

- $UU^\top/U_kU_k^\top$ is defined as a three-by-three orthogonal matrix whose entries are generated from normally distributed random variables. For $U_k^\top$, we simply generate three random U orthogonal matrices instead of generating one $UU^\top$ and applying to all three clusters.

4.1.1 Data Generation Function

For our simulation analysis, I built a function that outputs a data set fit to any of the fourteen possible Gaussian models represented in 4.1. This function was then input into another function I built that repeats this data generation process n number of times and clusters the data set through any of the fourteen Gaussian models.

What we are trying to understand from this simulation analysis is: what covariance models allow for us to recover clustering information as accurately as possible? In order for us to answer this question we need to choose models that express the uniqueness of each covariance parameter. For this experiment specifically, we will be using the VEE,EVE, and EEV covariance to create, and cluster the data.

For all of the simulations, there were basic properties that were constant across all experiments. So that the number of samples were robust and could easily be divided into three cluster groups, I set the number of observations to $n = 900$. Additionally, to get a robust output of adjusted Rand indices, I also set the number of repetitions to $r = 500$. Every model simulation was set $\mu_1 = (0, 0, 0)$, $\mu_2 = (5, 5, 5)$, $\mu_3 = (10, 10, 10)$. The reason behind these means being chosen for the analysis is so that when we run the clustering model, we are using a data set that does not make the clustering allocation too easy or too difficult. If the cluster means were too close to each other than cluster estimation will be too difficult. If the clusters are too far away from each other, than the cluster estimation will be easy. Lastly, for reproducibility, one seed was set for all of the simulations.

4.1.2 VEE Test

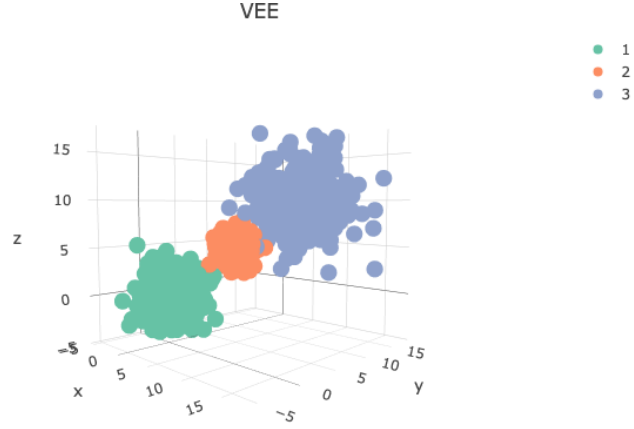


Figure 4.3: 3D plot of VEE generated data set

For the VEE model test, I defined $\lambda_1 = 6$ (intermediate volume cluster), $\lambda_2 = 2$ (small volume cluster), and $\lambda_3 = 14$ (large volume cluster). Additionally, $\Delta = \begin{pmatrix} 0.5 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 0.5 \end{pmatrix}$ (so that clusters aren't shaped in a non-spherical way). See Figure 4.3.

4.1.3 EVE Test

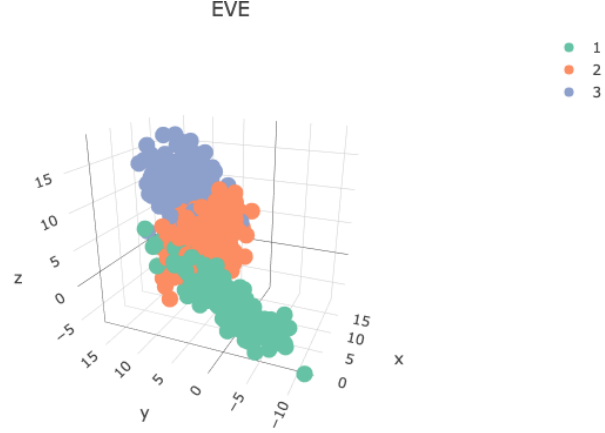


Figure 4.4: 3D plot of EVE generated data set

For the EVE model test, I defined $\lambda = (1, 1, 1)$ so that the size of the clusters are similar. Additionally, $\Delta_1 = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 17 \end{pmatrix}$ (yields a cluster shape that is more cigar shaped in the y-axis direction), $\Delta_2 = \begin{pmatrix} 6 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 3 \end{pmatrix}$ (yields a cluster shape that cigars in the x-axis and y-axis direction), and $\Delta_3 = \begin{pmatrix} 13 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 7 \end{pmatrix}$ (yields a cluster shape that cigars heavily in the x-axis direction and intermediately in the y-axis direction). See Figure 4.4.

4.1.4 EEV Test

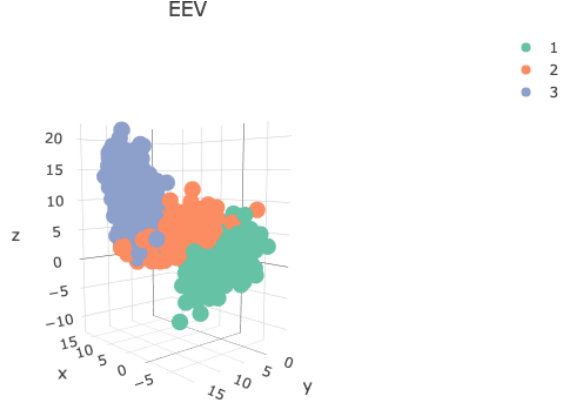


Figure 4.5: 3D plot of EEV generated data set

For the EEV model test. I defined $\lambda = (1, 1, 1)$ so that the size of the clusters are not similar. $\Delta_1 = \begin{pmatrix} 17 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}$, $\Delta_2 = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 15 & 0 \\ 0 & 0 & 3 \end{pmatrix}$, and $\Delta_3 = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 14 \end{pmatrix}$. The Δ_k are defined as such because we want to clearly define a “cigar” shape in each of the axis’ directions for each respective cluster). See Figure 4.5.

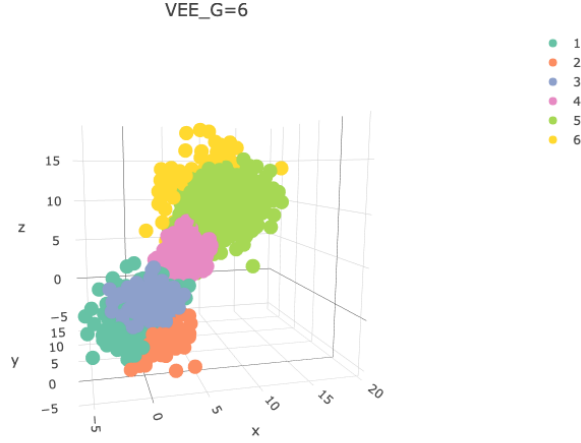


Figure 4.6: 3D plot of VVE generated data set with $G = 6$

For the clustering test. I kept the same parametric conditions as the VEE cluster test in Section 4.1.2, but adjusted $G = 3$ to now $G = 6$. See Figure 4.6.

Adjusted Rand Index: Determining the accuracy of a test can be an extremely subjective process. What does being accurate even mean? In this case, we want to assess how many data points were correctly allocated to a cluster after running each clustering scenario (VEE, EVE, EEV, and $G = 6$). For this, we turn to the Adjusted Rand Index metric.

Definition 4.1. The Adjusted Rand Index (ARI) is a statistical metric that quantifies the similarity between two partitions of a data set by comparing the assignments of data points to clusters. ARI is calculated by the following equation:

$$ARI(P^*, P) = \frac{\sum_{i,j} \binom{N_{ij}}{2} - \left[\sum_i \binom{N_i}{2} \sum_j \binom{N_j}{2} \right] / \binom{N}{2}}{\frac{1}{2} \left[\sum_i \binom{N_i}{2} + \sum_j \binom{N_j}{2} \right] - \left[\sum_i \binom{N_i}{2} \sum_j \binom{N_j}{2} \right] / \binom{N}{2}} \quad (4.2)$$

When $C^* = \text{class}$ and $C = \text{cluster}$, and N is defined as the number of data points in a data set and N_{ij} is the number of data points of the class label $C_j^* \in P^*$ assigned to cluster C_i in partition P . N_i is defined as the number of

data points in cluster C_i of partition P , and N_j is the number of data points in C_j^* . Generally, the ARI for a data set lies between 0 and 1. Where 1 indicates a clustering that is perfectly identical to the original data structure and 0 indicates random clustering.(Yang, 2017). ARI can also be a negative value indicating systematic difference between the two clusterings.

The final part of the simulation tests involves calculating the ARI for all 500 generated data sets for each scenario and derive observations. This test consists of developing three sets of 500 data sets under the following cases:

- Clustering done under the “correct” clustering model to serve as a **control** (EVE generated data set clustered under the EVE model).
- Clustering done under the **underfitting** “wrong” model (EVE generated data sets clustered under the EEE model).
- Clustering done under the **overfitting** “wrong” model (EVE generated data sets clustered under the VVV model).

These three cases will be run on data sets generated by the VEE, EVE, EEV, and VVE models (where $G = 6$ for the VVE model).

4.1.5 ARI Outcomes

The following figures will be of histograms displaying the ARIs of each clustering scenario.

VEE Test:

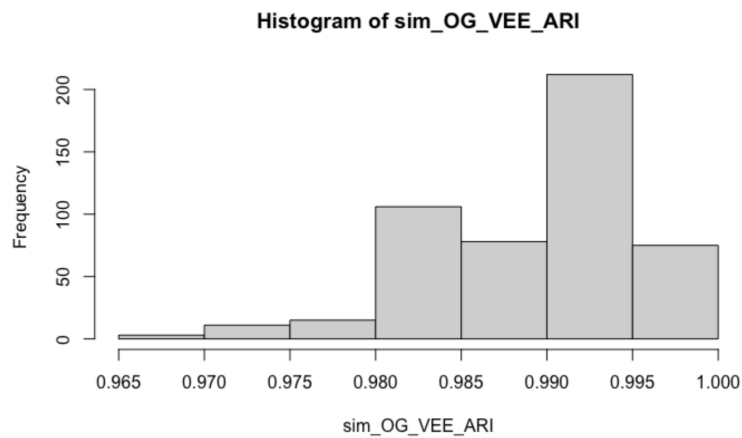


Figure 4.7: ARIs of VEE (correct) clustered VEE data sets

As expected, the ARIs for this test range from 0.965 to 1 indicating an extremely accurate clustering allocation. See Figure 4.7.

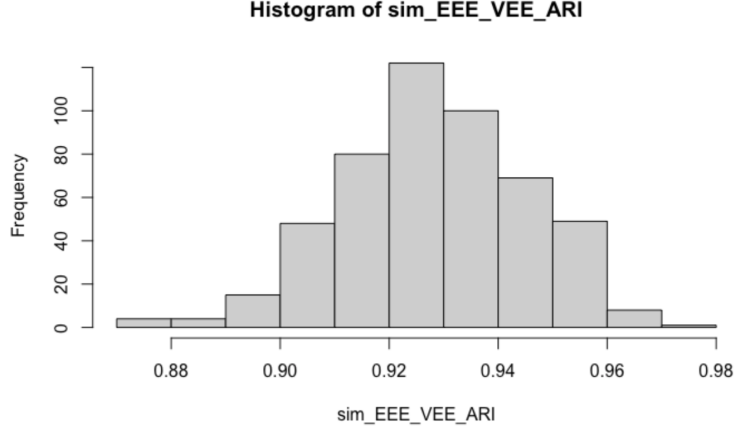


Figure 4.8: ARIs of EEE clustered (wrong) VEE data sets.

In 4.8, the underfitting case, we see the ARIs range from the lowest ARI of 0.88 to 0.98 with the highest frequency of data sets having an ARI of 0.93 indicating a high accuracy for this clustering test. See Figure 4.8.

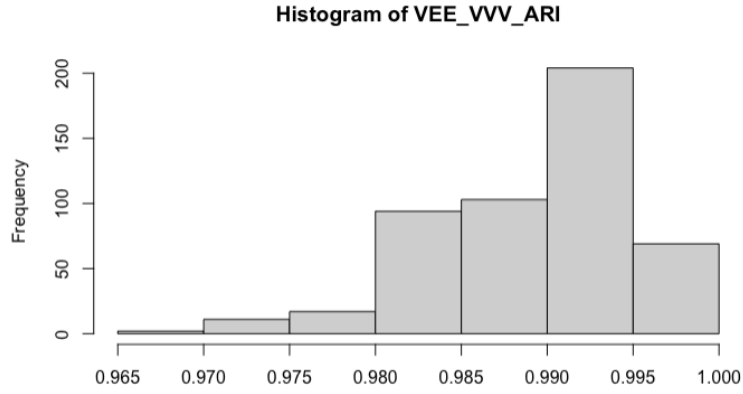


Figure 4.9: ARI distribution of VVV clustered (wrong) VEE data sets.

In Figure 4.9, the overfitting case, we see a left skewed shape, but more importantly, the range the ARIs of this test range from 0.965 to 1 indicating a high accuracy for this clustering test. VEE seems relatively easy to cluster. See Figure 4.9.

EVE Test:

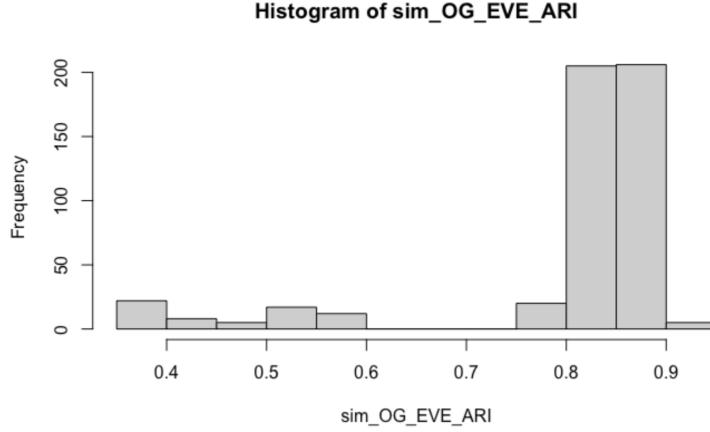


Figure 4.10: ARIs of EVE clustered (correct) EVE data sets.

In 4.10, the control case, we get yet another expected accurate output from our control test getting ARIs in the range of 0.8 to 0.9 for about 80% of our data sets. See Figure 4.10.

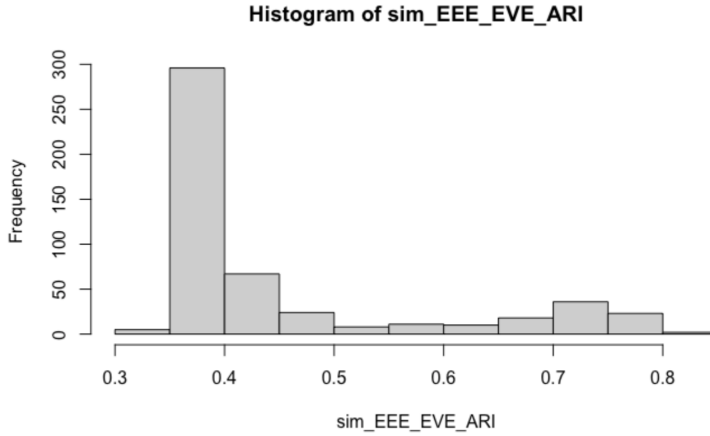


Figure 4.11: ARI distribution of EEE clustered (wrong) EVE data sets.

In Figure 4.11, the underfitting case, we see an explicit peak form at 0.4

ARI with approximately 60% of the observations having an $\text{ARI} > 0.4$. The clustering accuracy was low for this case. See Figure 4.11.

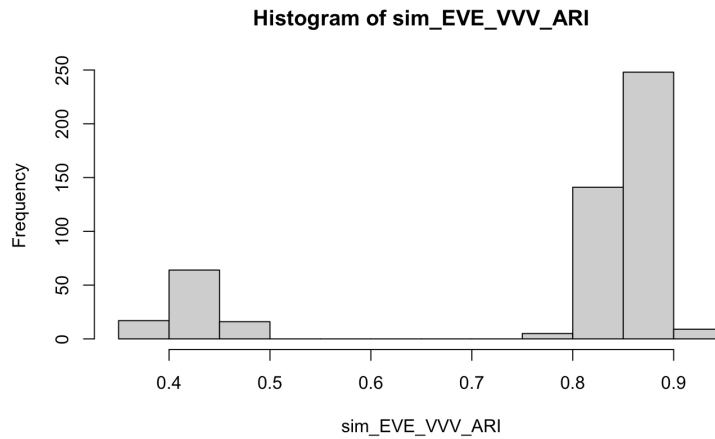


Figure 4.12: ARIs of VVV clustered (wrong) EVE data sets.

In Figure 4.12, the overfitting case, we also get relatively high accuracy results. Although in this case, we have a grouping (at most 20% of the observations) of ARIs that fall < 0.6 . But overall, the vast majority of the data sets fell within a desirable ARI range of 0.75 and higher. See Figure 4.12.

EEV Test:

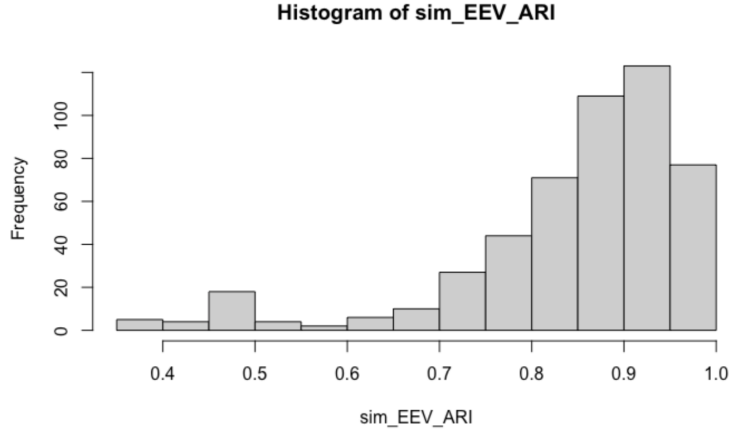


Figure 4.13: ARI distribution of EEV clustered (correct) EEV data sets.

In Figure 4.13, the control case, we get a somewhat expected output (higher density in the higher ARI values) with an unexpected (left skew) distribution. See Figure 4.13.

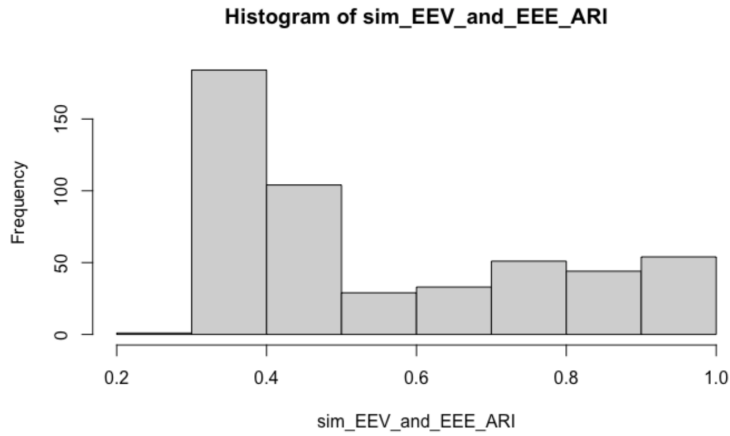


Figure 4.14: ARI distribution of EEE clustered (wrong) VEE data sets.

In Figure 4.14, the underfitting case, we can see a right skewed distribution of ARIs that peaks at 0.4 having approximately 90% of the ARIs < 0.5 . See Figure 4.14.

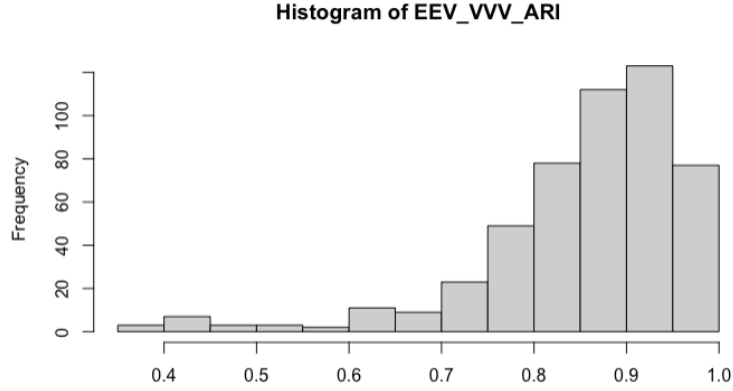


Figure 4.15: ARI distribution of VVV clustered (wrong) VEE data sets.

In Figure 4.15, the overfitting case, we get a very similar result from our control case: a left skewed ARI distribution, indicating good performance. See Figure 4.15.

G = 6 Clusters Test:

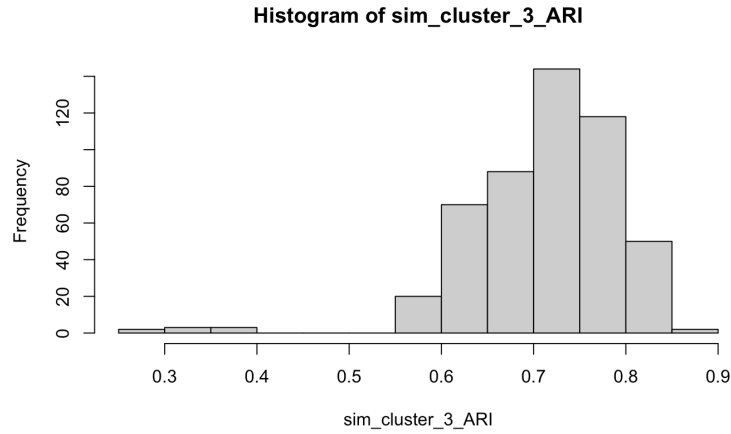


Figure 4.16: ARI distribution of VVE data set where $G = 3$ clustered under the VVE model.

In Figure 4.16, the control case, we get a majority proportion of ARIs ranging from 0.55 to 0.95 peaking at the 0.75 ARI value. See Figure 4.16.

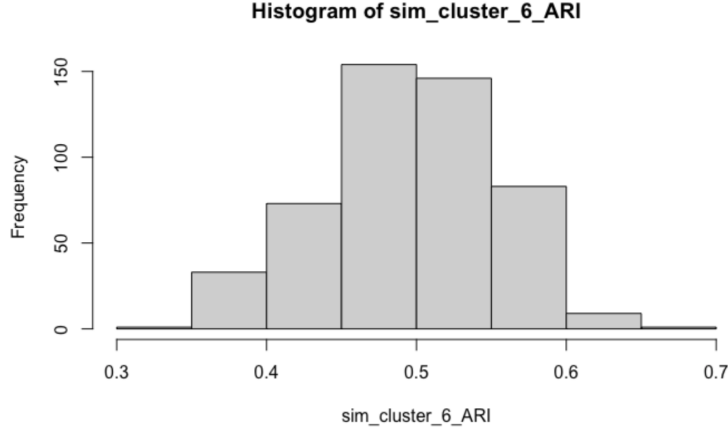


Figure 4.17: ARI distribution of VVE data set where $G = 6$ clustered under the VVE model.

In Figure 4.17, the underfitting case, we get ARI outputs that are roughly normally distributed peaking at 0.5 ARI. See Figure 4.17.

G = 6 Clusters and EEE Model Test:

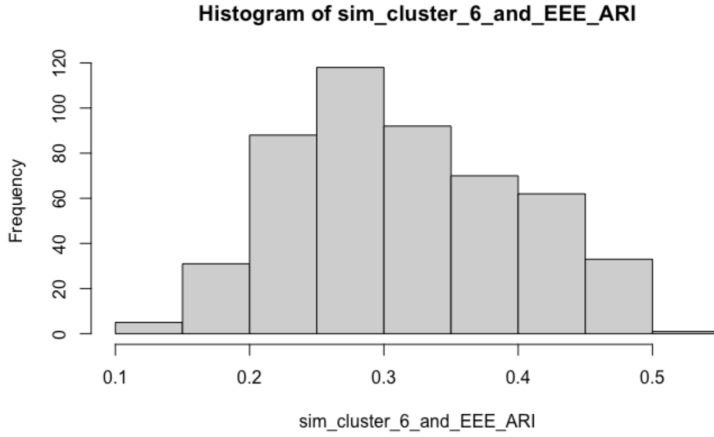


Figure 4.18: ARI distribution of VVE data set clustered under the EEE model and $G = 6$.

In Figure 4.18, we get ARI outputs that are roughly normally distributed

peaking at 0.3 indicating a low accuracy for this test.

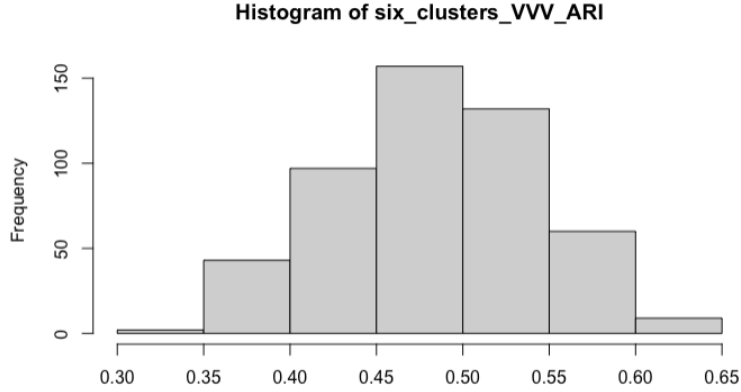


Figure 4.19: ARI distribution of VVE data set clustered under the VVV (wrong) model and $G = 6$.

In Figure 4.19, we get ARI outputs that are roughly normally distributed peaking at 0.5 indicating a low accuracy for this test.

4.1.6 ARI Distribution Inference

VEE Inference: One element of this test that stands out to me is the overall high accuracy outputs from the VEE. Starting with Figure 4.8, this underfitting analysis yielded high ARIs ranging from 0.86 to 0.98. This could potentially indicate the model being more capable to cluster data that adopts a more volumetrically varying distribution. Taking this a step further in Figure 4.9, we see a distribution of even higher ARIs ranging from 0.965 to 1. This could indicate VVV being a strong choice when choosing a clustering method that adopts a more volumetrically varying distribution.

EVE Inference: One of the properties that I recognize from these EVE tests is the stark contrast in ARI outputs from the EEE clustering and the VVV clustering. The EEE clustering test yielded our lowest high density range of ARIs. Underfitting the data output around 70% of our data sets in the 0.35 to 0.45 ARI range. On the other hand, our VVV overfitting test yielded a relatively high percentage (80%) of data sets falling within the 0.8 to 0.9 ARI range. This test on EVE distributed data sets could

be another piece of evidence that leads us to believe overfitting as a more accurate approach when clustering. A question that was raised for me while analyzing these figures is how there was a relatively small proportion of data sets that served almost as outliers in each test. For the control, a small subset of data sets output ARIs less than 0.6, contrary to the majority of data sets falling within the greater than 0.75 ARI range. Additionally for the underfitting test there was a small proportion did not fall in the 0.35 to 0.45 ARI range. This “outlier” subset were surprisingly clustered accurately, ranging from 0.65 to 0.8. And finally for the overfitting test there was a relatively small proportion (20%) of data sets that output ARIs less than 0.5.

EEV Inference: The EEV test yielded some of the most unique results yet. For both the control test and the VVV test, the ARIs adopted a left skewed distribution both peaking at 0.9 ARI. Additionally, the VVV ARI distributions following a very similar distribution to the control could also be a sign showing overfitting as the better clustering approach. With the underfitting test, we see the distribution peak around the 0.4 ARI area. This observation could be more evidence of underfitting the data as an unsuitable approach when clustering data.

G=6 Clusters Inference: Something about this test that stands out from the rest of them is the parametric estimation increase. Referring back to Figure 4.2, parameter estimation for non VEE, EVE, and EEV models increases significantly. As a result, we see a significant decrease in ARI distribution between the $G = 6$ clustering control test and the other control tests that were done. In the control case, the data sets output ARIs that roughly distributes normally around 0.75. As seen in Figure 4.17, the ARIs distribute into a normal distribution with a peak at 0.5 indicating a weak clustering performance for this specific case. I think because there were so many factors to take into account (number of clusters and more parameters to estimate) this specific case yielded much lower ARIs. Additionally, the low ARIs occur because $G = 3$ partitions being clustered into $G = 6$ clusters will never yield a perfect alignment. be a result of the

G=6 Clusters and EEE Inference: For this final analysis, we get very similar results to the prior $G = 6$ clusters. For the underfit modeling test, the ARIs distribute normally with a peak at 0.3. For the overfit test in

Figure 4.18, we also see a normal distribution of ARIs centered at 0.475.

Final Remarks: Overfitting the data proved to be one of the more reliable approaches throughout the various model tests. Recognizing the similarity in ARI distributions between the control and overfitting tests in Figure 4.10/Figure 4.12 and in Figure 4.13/Figure 4.12 contributes heavily to this hypothesis. Even though the process of estimating more parameters takes more computationally energy and potentially result in overfitting, the accuracy it provides could outweigh the downfalls. Choosing a model that does not estimate as many parameters as the VVV model, but still requires computational rigor could be a valid approach to data clustering.

Chapter 5

Application: Analyzing Swing Characteristics through Gaussian Mixture Models

With the release of new bat tracking metrics in the MLB, the world of baseball analytics has opened up the door to whole new world. As described in Chapter 2, these five new metrics have given enthusiasts of baseball analytics a mathematical way of describing and analyzing their favorite player's swing characteristics. A prime example of this idea would be in describing a player like Luis Arraez. Previously, people would characterize Luis Arraez as a contact-focused batter who rarely attempts long and strong swings. How does he do this? Arraez limits strikeouts by limiting big boisterous swings instead opting to take short swings to limit room for error. But how do we know what a short swing is? With the new metrics, we can observe that in 2025, his average swing length was measured at 5.7 feet which in comparison to a prolific home run hitter like Shohei Ohtani whose average bat length in 2025 was measured at 7.9 feet. Additionally, Arraez's average bat speed measured in at 62.6 miles per hour which ranks in the bottom 1 percentile (BaseballSavant, 2024). All of that to say, being able to associate quantifiable metrics to the qualitative observations analysts make allow us to compare and evaluate players in a whole new manner.

To open up the door to these new metrics, we will be investigating a player that has performed at the highest level for over a decade in the MLB: Jose Altuve. Within our analysis, we will only be using registered swing attempts

(a swing attempt is classified as any event that resulted in either a swinging strikeout, foul tip into glove, foul ball, or hit into play). Additionally, for our clustering analysis, I decided to let $G = 3$ and used the VVV clustering model (to be in accordance to my analysis in Chapter 4). The situations we want to analyze would include key factors that may alter a batters swing metrics. *The following situations we will analyze are:

- **Two-strike Vs. Non-Two Strike Situations-** This situation is one of the most well known swing altering situations. For example, to prevent a strikeout from occurring, a batter may adjust their bat speed or swing length to prevent a strikeout. This concept is often called having a “two-strike approach”.
- **Varying Pitch Type Situations-** Different pitch types can alter a batters swing in many different ways. A fastball can make a batter increase their bat speed to so that they can make contact with the ball before it speeds past them, a curveball with downwards movement can make a batter increase their swing path tilt to reach the falling ball, etc.
- **Varying Pitch Location Situations-** This situation is very important as well because it encompasses every situation a batter will need to alter their swing to make contact with the baseball. Additionally, inference can be made on what areas of the strike zone a batter is more comfortable with.

For our analysis, I took 2025 bat tracking data from the Houston Astro’s Second Baseman Jose Altuve. I then filtered down to pitches seen that resulted in a swing (meaning the result of the pitch being either a foul ball, foul tip into glove, swinging strike, groundout, flyout, lineout, or hit). I then clustered the data using the VVV covariance model with $G = 3$. For the test ran in Figure 5.1, I grouped the data into two bins: swings occurring when there are less than 2 strikes, and swings occurring when there are two strikes. Finally, we calculated the summary squared up percentage and contact rate for each cluster. For the test conducted in Figure 5.2, I followed same process used for Figure 5.1 accept grouped the data by pitch type after clustering. *Same process was also used for Figure 5.3 accept the data is grouped by pitch location after clustering.

Note: To allow for seamless analysis of all five bat tracking metrics, I normalized the data. Additionally, I used a swing metrics data that contained 1136 swing attempts spanning the entire 2025 MLB season.

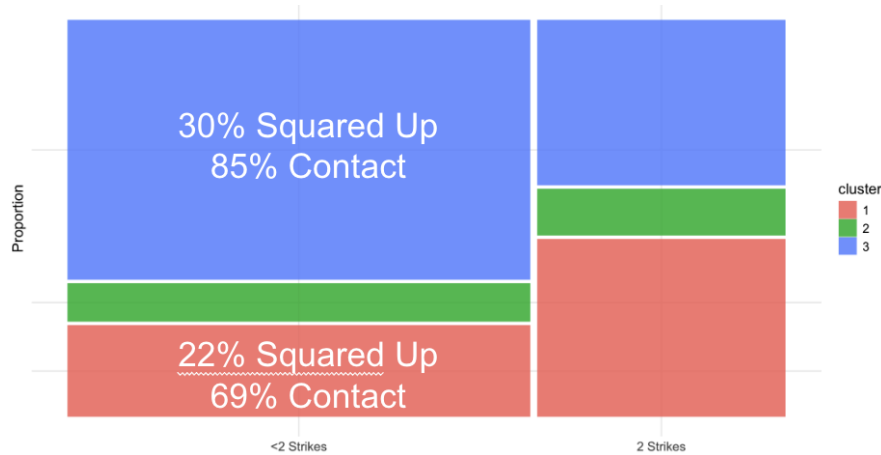


Figure 5.1: Jose Altuve two-strike and non two-strike clustered swing attempts.

In Figure 5.1 we get a distribution of swings that distinctly fit two-different swing outcomes. For Cluster 3, we see it characterized with having a substantially higher squared up percentage in comparison to Cluster 1 whose average squared up percentage comes in 8% lower than Cluster 3. Additionally the contact rate (contact rate is the percentage of times a batter makes contact with the baseball) in Cluster 3 is 16% higher than Cluster 1’s contact rate. One way we can interpret Figure 5.1 is by associating the Cluster 3 swings as a more comfortable, confident “A tier” swing whereas swings falling in Cluster 1 would be denoted as being more of a “B tier” swing that was taken out of discomfort. Lastly, we can see a substantially larger proportion of these Cluster 3 swings happening when the strike count is less than 2. For 2 strike situations, we see the inverse occur: a substantially larger proportion of Cluster 1 swings occur compared to the Cluster 1 proportion in the less than two-strike situation.

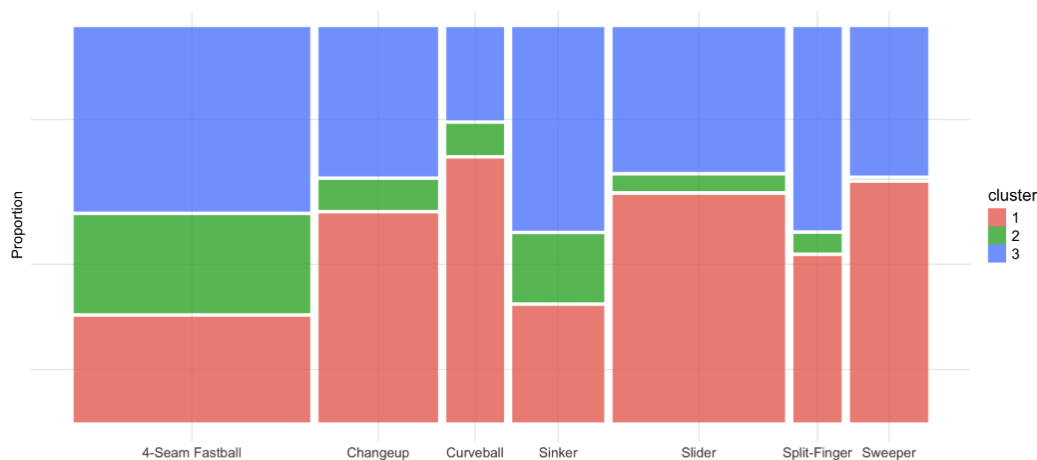


Figure 5.2: Jose Altuve clustered swing attempts by pitch type.

In Figure 5.2, we get a vast distribution of different swing clusters. One thing that stands out to me is the proportion of cluster 3 swings being higher in fastball pitches (4-seam fastball, split-finger fastballs, and sinkers) compared to the off-speed/breaking pitches (pitches that are characterized by having either horizontal movement or downwards movement). All three fastball pitches have the highest proportion of cluster 3 swings than any other pitch type. Additionally, all of the off speed pitches contain the highest proportion of cluster 1 swings. This could indicate Altuve being a more successful hitter when presented with a fastball variant compared to when he is presented with an off speed pitch type.

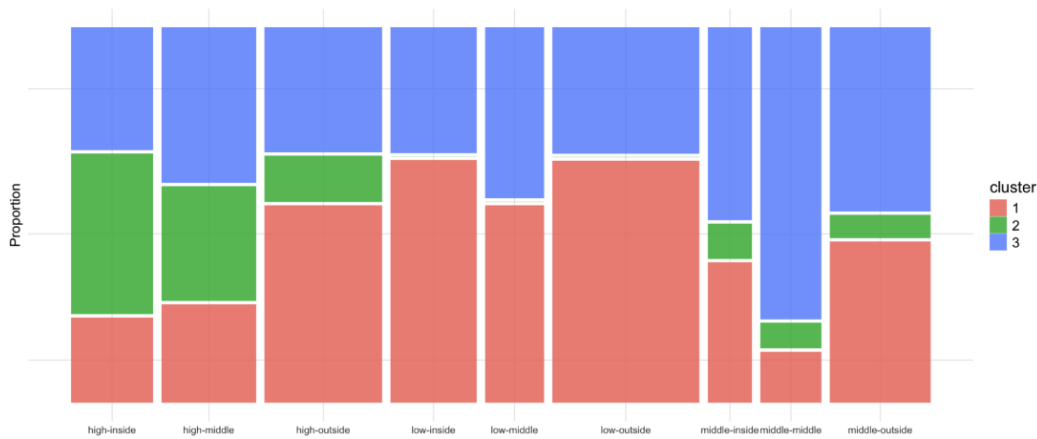


Figure 5.3: Jose Altuve clustered swing attempts by pitch location.

Figure 5.3 yields a very interesting output. The pitch locations with the highest proportion of cluster 3 are pitches that are located in the middle of the strike zone. This result makes sense given that the center of the strike zone would not require a batter to alter their body in an uncomfortable way. The second highest proportion of cluster 3 swings resides in the middle-inside sector of the strikeout, potentially displaying some comfort with pitches being thrown inside as well. Most notably, Altuve's highest proportion of cluster 1 swings occurred when swinging at pitches located low and on the outside of the strike zone, hinting at him making uncomfortable swings when presented with pitches in that location.

Chapter 6

Conclusion

Within this thesis, we set out to explore the Gaussian mixture model and investigate how it could be applied to MLB bat tracking data to examine whether batters possess statistically distinguishable swing types. To answer this question, we approached the problem in three stages: establishing the theoretical foundations of mixture models and the EM algorithm, testing the behavior of different covariance structures through simulation, and applying the methodology to a real player’s swing data.

We began laying the groundwork by introducing mixture distributions and deriving the EM algorithm. This framing is essential for the baseball application, as a batter’s swing data can be interpreted as a sample drawn from multiple underlying swing types. Working through the derivations of parameters helped show how the EM algorithm iteratively recovers the parameters that maximize the likelihood. Understanding how the E-step and M-step alternate gives us a concrete sense of what a Gaussian mixture model is actually doing when it clusters a data set.

We explored how the 14 possible Gaussian models differ in their treatment of volume, shape, and orientation. The simulation study was designed around a simple question: when we do not know the true underlying structure of a data set, which model should we choose? Across the VEE, EVE, and EEV tests, overfitting the data with the VVV model tended to produce ARI distributions that closely resembled the control cases, while underfitting with the EEE model tended to produce noticeably lower ARIs. This result is somewhat counterintuitive, because fitting more parameters than

necessary could be expected to introduce noise. However, within the range of scenarios we tested, the flexibility of the VVV model seemed to help more than it hurt. The $G = 6$ test complicated this picture, as once the number of clusters increased, even the correctly specified model saw its ARIs drop considerably, due to the fact that we had the “wrong” number of clusters. This suggests that the benefits of flexibility have limits when the estimation problem becomes more complex.

For our bat tracking analysis, we applied the VVV model with $G = 3$ to 1,136 of Jose Altuve’s registered swing attempts from the 2025 MLB season. The three situational breakdowns (two-strike counts, pitch type, and pitch location) each produced patterns that are consistent with how analysts have described Altuve qualitatively. Cluster 3 swings, which had higher squared-up and contact rates, appeared more often in non-two-strike counts, against fastballs, and on pitches in the middle of the strike zone. Cluster 1 swings, which had lower squared-up and contact rates, appeared more often in two-strike counts, against off-speed and breaking pitches, and on pitches low and outside. It is important not to overstate what this shows. Three clusters is a modeling choice rather than a discovery, and the labels “A tier” and “B tier” are interpretations rather than ground truth. However, the fact that the clusters sorted themselves cleanly across situational variables that baseball analysts care about is encouraging evidence that the approach is picking up on something real.

Returning to the question raised in Chapter 1, whether batters actually possess multiple swings as Bryce Harper described, this analysis suggests that at least for Jose Altuve, the data supports the idea that more than one swing is being deployed, and that the selection between swings is responsive to the game situation. Whether this generalizes across players, and whether three is the correct number of swing types to look for, are questions that were not answered within the scope of this thesis.

There are a number of limitations and directions for future work that are worth acknowledging. First, we chose $G = 3$ partly for interpretability and partly because it produced clean results, but we did not use a formal model selection criterion like BIC to justify this choice. A more rigorous extension of this work would allow the data to inform the appropriate number of clusters. Second, this analysis focused entirely on one player, so applying the

same framework across many hitters could reveal whether there are league-wide swing archetypes or whether each player's clustering is truly individual. Third, the bat tracking metrics released by Baseball Savant are still relatively new, and the set of available features will likely expand in coming seasons, which could allow for richer clustering in higher-dimensional spaces. Additionally, since the data spans only one full MLB season, there is definitely room to grow this analysis once more data is created.

Baseball analytics has spent the last two decades closing the gap between what scouts see and what the data can measure. While this thesis is a small contribution to that effort, it demonstrates that Gaussian mixture models provide a useful framework for translating the qualitative language of baseball scouting into quantifiable structure. Where an analyst might previously have said that Altuve looks uncomfortable on pitches low and away, we can now show that such swings cluster separately from his preferred swing and differ measurably along the five bat tracking dimensions. The swings in a hitter's arsenal, once describable only through metaphor, are now at least partially recoverable as clusters in a five-dimensional space.

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Technology Statement

Data creation and analysis was conducted solely through RStudio. The following packages within RStudio were used in my thesis tidyverse, ggplot2, mclust, ggrepel, greed, GGally, dplyr, flexmix, MASS, mvtnorm, JWileymisc, plotly, purrr, and sabRmetrics. Generative AI was not used in my thesis.

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